DOMINICK BRAICO

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Education

University of Michigan, M.S. Robotics — Incoming August 2026

University of Illinois Urbana-Champaign, B.S. Mechanical Engineering, Minor in Computer Science — May 2026
Relevant Coursework: Software Development for Mobile Robots, Data Structures and Algorithms, Machine Learning, Robot Dynamics, Dynamics of Mechanical Systems, Signal Processing, Machine Design I/II, Electronic Circuits, CAD

Work Experience

Powertrain Engineering Intern — May 2025 – Aug 2025
Whisper Aero
- Architected a Python GUI over CAN bus for real-time parameter tuning and telemetry of high-power density 3-phase motor controllers, applying object-oriented design patterns for modular instrument control
- Automated dynamometer test sequences in Python, building structured data pipelines that translated raw sensor streams into structured engineering datasets and reports
- Developed C/C++ firmware for FOC-based motor controllers, implementing low-level control logic interfacing directly with hardware registers and sensor feedback loops

Field Robotics Engineering Intern — May 2024 – March 2025
Robotics for Engineer Operations - Construction Engineering Research Laboratory
- Validated software and hardware integration across mechanical, electrical, and autonomy subsystems on Co3MaNDR, a modular construction robot, diagnosing cross-system failures and translating between sensor data formats and inputs
- Designed a rugged electronic control module for robot sensors and compute unit, reducing electrical system volume and improving weatherproofing on a modular cable-driven construction robot

Extracurriculars & Leadership

Control Systems & Sensor Integration Lead — February 2024 - Present
GHOST Electric Motorcycles
- Configured CAN bus network for communication between motor controller, BMS, and onboard sensors; performed bench testing to validate electrical system performance prior to vehicle integration
- Implemented feedback control algorithms to tune torque response of a 45kW PMAC motor
- Standardized wiring harness and high-power cabling for a 103.6V nominal voltage electric motorcycle

Projects

ASNE PEP Autonomous Surface Vessel Competition
- Designed and validated a full-stack autonomous navigation suite fusing GPS and IMU data for high-precision waypoint maneuvering on an autonomous surface vessel (ASV)
- Built a real-time object detection pipeline using OpenCV on a raspberry pi to identify and classify racecourse buoys for collision avoidance and course navigation
- Integrated high-performance electric propulsion systems and motor controllers through field testing, optimizing vessel performance against mission-specific operational requirements

Robotic Mobile Manipulator Simulation
- Developed custom URDF files and Gazebo world configurations to build a high-fidelity simulation environment integrating a Clearpath Husky mobile base, UR3 manipulator arm, and Robotiq 2F-85 gripper, defining robot kinematics, joint constraints, and sensor placement entirely in software
- Authored ROS 2 launch files and controller configuration files to orchestrate multi-node system startup, parameter loading, and hardware interface bridging between simulated and real robot controllers
- Implemented SLAM using LiDAR point cloud data and the ROS 2 navigation stack, writing Python nodes that translated raw sensor streams into occupancy grid maps for real-time localization and path planning
- Integrated virtual sensor systems including wheel encoders and LiDAR for state estimation, developing modular Python-based ROS 2 nodes for data acquisition and sensor fusion

Skills

Mechanical Design: SolidWorks (Certified Mechanical Design Associate), Autodesk Fusion
Fabrication: 3D Printing, Waterjet, Laser Cutter, Shop Tools, Soldering
Programming: Python, C/C++, ROS 2